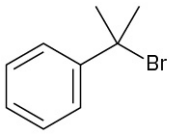
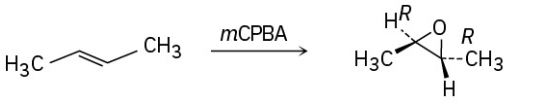
PROBLEM 18-7 **(a)**  **(b)** 

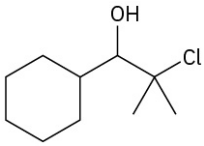
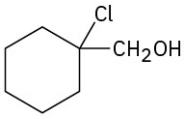
PROBLEM 18-8 Protonation of the oxygen atom, followed by E1 reaction

PROBLEM 18-9 Br^- and I^- are better nucleophiles than Cl^-.

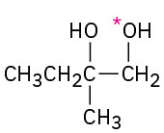
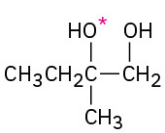
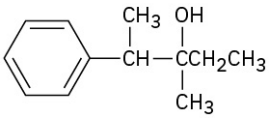
PROBLEM 18-10 

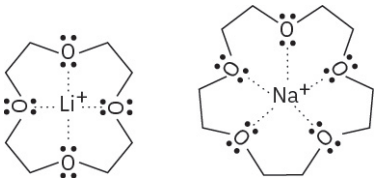
trans-2-Butene trans-2,3-Epoxybutane

PROBLEM 18-11 Epoxidation of *cis*-2-butene yields *cis*-2,3-epoxybutane, while epoxidation of *trans*-2-butene yields *trans*-2,3-epoxybutane.

PROBLEM 18-12 **(a)**  **(b)** 

PROBLEM 18-13 **(a)** 1-Methylcyclohexene + OsO_4; then $NaHSO_3$ **(b)** 1-Methylcyclohexene + *m*-chloroperoxybenzoic acid, then H_3O^+

PROBLEM 18-14 **(a)**  **(b)**  **(c)** 

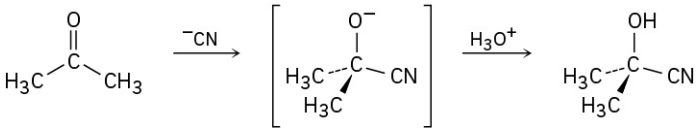
PROBLEM 18-15 

PROBLEM 18-16 **(a)** 2-Butanethiol **(b)** 2,2,6-Trimethyl-4-heptanethiol **(c)** 2-Cyclopentene-1-thiol **(d)** Ethyl isopropyl sulfide **(e)** *o*-Di(methylthio)benzene **(f)** 3-(Ethylthio)cyclohexanone

PROBLEM 18-17 **(a)** 1. $LiAlH_4$; 2. PBr_3; 3. $(H_2N)_2C{=}S$; 4. H_2O, NaOH **(b)** 1. HBr; 2. $(H_2N)_2C{=}S$; 3. H_2O, NaOH

PROBLEM 18-18 1,2-Epoxybutane

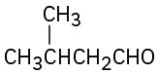
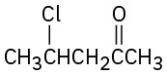
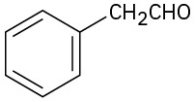
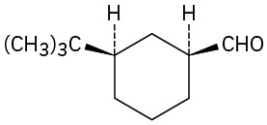
PROBLEM 18-69 Acetyl chloride is more electrophilic than acetone.

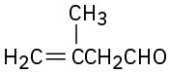
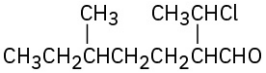
PROBLEM 18-70 

PROBLEM 18-71 **(a)** Nucleophilic acyl substitution **(b)** Nucleophilic addition **(c)** Carbonyl condensation

Chapter 19

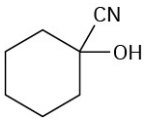
PROBLEM 19-1 **(a)** 2-Methyl-3-pentanone **(b)** 3-Phenylpropanal **(c)** 2,6-Octanedione **(d)** *trans*-2-Methylcyclohexanecarbaldehyde **(e)** 4-Hexenal **(f)** *cis*-2,5-Dimethylcyclohexanone

PROBLEM 19-2 **(a)**  **(b)**  **(c)**  **(d)** 

(e)  **(f)** 

PROBLEM 19-3 **(a)** Dess–Martin periodinane **(b)** 1. O_3; 2. Zn **(c)** DIBAH **(d)** 1. BH_3, then H_2O_2, NaOH; 2. Dess–Martin periodinane

PROBLEM 19-4 **(a)** $HgSO_4$, H_3O^+ **(b)** 1. CH_3COCl, $AlCl_3$; 2. Br_2, $FeBr_3$ **(c)** 1. Mg; 2. CH_3CHO; 3. H_3O^+; 4. CrO_3 **(d)** 1. BH_3; 2. H_2O_2, NaOH; 3. CrO_3

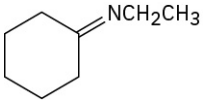
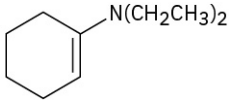
PROBLEM 19-5 

PROBLEM 19-6 The electron-withdrawing nitro group in *p*-nitrobenzaldehyde polarizes the carbonyl group.

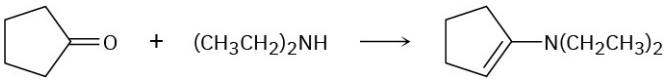
PROBLEM 19-7 $CCl_3CH(OH)_2$

PROBLEM 19-8 Labeled water adds reversibly to the carbonyl group.

PROBLEM 19-9 The equilibrium is unfavorable for sterically hindered ketones.

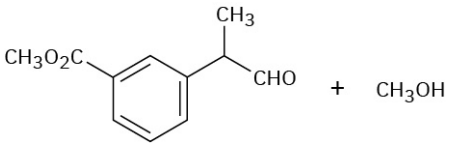
PROBLEM 19-10  and 

PROBLEM 19-11 The steps are the exact reverse of the forward reaction shown in Figure 19.7.

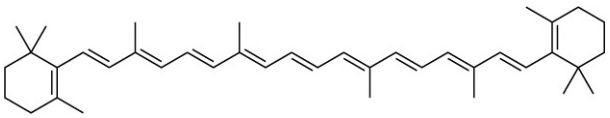
PROBLEM 19-12 

PROBLEM 19-13 **(a)** H_2/Pd **(b)** N_2H_4, KOH **(c)** 1. H_2/Pd; 2. N_2H_4, KOH

PROBLEM 19-14 The mechanism is identical to that between a ketone and 2 equivalents of a monoalcohol, shown in Figure 19.11.

PROBLEM 19-15 

PROBLEM 19-16 **(a)** Cyclohexanone + $(Ph)_3P{=}CHCH_3$ **(b)** Cyclohexanecarbaldehyde + $(Ph)_3P{=}CH_2$ **(c)** Acetone + $(Ph)_3P{=}CHCH_2CH_2CH_3$ **(d)** Acetone + $(Ph)_3P{=}CHPh$ **(e)** $PhCOCH_3$ + $(Ph)_3P{=}CHPh$ **(f)** 2-Cyclohexenone + $(Ph)_3P{=}CH_2$

PROBLEM 19-17 

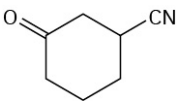
***β*-Carotene**

PROBLEM 19-18 Intramolecular Cannizzaro reaction

PROBLEM 19-19 Addition of the pro-*R* hydrogen of NADH takes place on the Re face of pyruvate.

PROBLEM 19-20 The –OH group adds to the Re face at C2, and –H adds to the Re face at C3, to yield (2*R*,3*S*)-isocitrate.

PROBLEM 19-21



PROBLEM 19-22 **(a)** 3-Buten-2-one + $(CH_3CH_2CH_2)_2CuLi$ **(b)** 3-Methyl-2-cyclohexenone + $(CH_3)_2CuLi$ **(c)** 4-*tert*-Butyl-2-cyclohexenone + $(CH_3CH_2)_2CuLi$ **(d)** Unsaturated ketone + $(H_2C{=}CH)_2CuLi$

PROBLEM 19-23 Look for appearance of either an alcohol or a saturated ketone in the product.

PROBLEM 19-24 **(a)** 1715 cm^{-1} **(b)** 1685 cm^{-1} **(c)** 1750 cm^{-1} **(d)** 1705 cm^{-1} **(e)** 1715 cm^{-1} **(f)** 1705 cm^{-1}

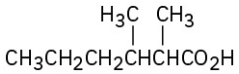
PROBLEM 19-25 **(a)** Different peaks due to McLafferty rearrangement **(b)** Different peaks due to α cleavage and McLafferty rearrangement **(c)** Different peaks due to McLafferty rearrangement

PROBLEM 19-26 IR: 1750 cm^{-1}; MS: 140, 84

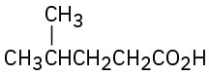
Chapter 20

PROBLEM 20-1 **(a)** 3-Methylbutanoic acid **(b)** 4-Bromopentanoic acid **(c)** 2-Ethylpentanoic acid **(d)** *cis*-4-Hexenoic acid **(e)** 2,4-Dimethylpentanenitrile **(f)** *cis*-1,3-Cyclopentanedicarboxylic acid

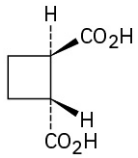
PROBLEM 20-2 **(a)**



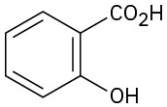
(b)



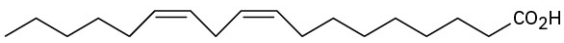
(c)



(d)



(e)



(f) $CH_3CH_2CH{=}CHCN$

PROBLEM 20-3 Dissolve the mixture in ether, extract with aqueous NaOH, separate and acidify the aqueous layer, and extract with ether.

PROBLEM 20-4 43%

PROBLEM 20-5 **(a)** 82% dissociation **(b)** 73% dissociation

PROBLEM 20-6 Lactic acid is stronger because of the inductive effect of the –OH group.

PROBLEM 20-7 The dianion is destabilized by repulsion between charges.

PROBLEM 20-8 More reactive

PROBLEM 20-9 **(a)** *p*-Methylbenzoic acid < Benzoic acid < *p*-Chlorobenzoic acid **(b)** Acetic acid < Benzoic acid < *p*-Nitrobenzoic acid

PROBLEM 20-10 **(a)** 1. Mg; 2. CO_2; 3. H_3O^+ **(b)** 1. Mg; 2. CO_2; 3. H_3O^+ or 1. NaCN; 2. H_3O^+

PROBLEM 20-11 1. NaCN; 2. H_3O^+; 3. $LiAlH_4$

PROBLEM 20-12 1. PBr_3; 2. NaCN; 3. H_3O^+; 4. $LiAlH_4$

PROBLEM 20-13 **(a)** Propanenitrile + CH_3CH_2MgBr, then H_3O^+ **(b)** *p*-Chlorobenzonitrile + CH_3MgBr, then H_3O^+

PROBLEM 20-14 1. NaCN; 2. CH_3CH_2MgBr, then H_3O^+